

Capstone Project

Project Title:

E-Commerce End-to-End Automation Testing (OpenCart)

Project Overview:

This project aims to design and implement a comprehensive, automated Test suite for the OpenCart e-commerce demonstration platform. The framework will validate 8 core business services with a minimum of 120 test cases, ensuring high software quality, functional stability, and seamless user experience across the application.

Target Application:

Website: Naveen automation labs

URL: <https://naveenautomationlabs.com/opencart/>

Technology Stack:

Playwright, JavaScript, Node.js

Services:

1. Authentication Service
2. Search Service
3. Product Service
4. Cart Service
5. Checkout Service
6. User Profile Service
7. Wishlist Service
8. Order Management Service

Project Deliverables:

- Playwright test automation codebase hosted on Git.
- Minimum 120 automated test scripts spanning 8 core services.
- Execution instructions (README.md).
- Allure execution report showing pass/fail metrics.

Conclusion

The completion of this capstone project demonstrates the power and efficiency of Playwright with JavaScript for modern web automation. By testing 8 distinct services and implementing the Page Object Model (POM), this framework ensures the stability of the OpenCart platform. Furthermore, the integration of Allure Reports bridges the gap between academic training and enterprise-level QA reporting, providing a scalable, reliable solution for real-world software testing.